Mutual Incoherence and Relative Total Variation Regularizations for Blind Hyperspectral Unmixing

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***Abstract*— The unsupervised hyperspectral unmixing (HU) technique, also known as blind HU, directly decomposes the mixed pixels of a hyperspectral image (HSI) into a combination of endmembers weighted by their fractional abundances, without relying on a spectral library. To perform the blind HU, nonnegative matrix factorization (NMF) and its extensions have been widely used to improve the performance of the abundances, with sparse constraints and spatial information about the abundances. However, there are only a few works that exploit the prior information or constraints on the endmembers in the NMF-based methods for the blind HU. The linear mixing model (LMM) assumption assumes that the endmembers should be uncorrelated. Unfortunately, the spectra of HSI are highly correlated in practice, making it challenging to decompose the mixed spectra using only spatial information. Therefore, it is necessary to consider the proper constraints on the endmembers to obtain endmembers with higher accuracy from HSI. This article first introduces the mutual incoherence (MIC) regularization to promote incoherence between endmembers, that is, to minimize the correlation between the estimated endmembers. Then, the relative total variation (RTV) regularization is introduced on the abundances to exploit the spatial structure of the HSI. Finally, the MRNMF algorithm based on NMF is proposed for the HU task using multiplicative iteration rule. The experimental results on synthetic and real datasets show that the proposed MRNMF outperforms the state-of-the-art blind HU algorithms.**

***Index Terms*— Hyperspectral unmixing (HU), mutual incoherence (MIC), nonnegative matrix factorization (NMF), relative total variation (RTV).**

I. Introduction

IN REMOTE sensing, the emergence and rapid development of hyperspectral images (HSIs) have resulted in significant changes in the theory and applications of remote sensing technology. By carrying different sensors, the hyperspectral technology can enable the target region to be imaged simultaneously from tens to hundreds of continuous spectral bands, which makes the HSIs contain rich spectral information. Therefore, the HSIs have been applied in many fields, such as space exploration, land use analysis, mineral detection, environmental monitoring, agriculture, and military applications [1], [2], [3], [4], [5].

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However, due to the spatial resolution of HSIs and the highly complex distribution of objects in the real-world scene, the pixel of the HSIs usually contains the mixed spectral information of several materials and is also named the mixed pixel [6]. The existence of mixed pixels seriously degrades the performance of HSIs applications [7], and hyperspectral unmixing (HU) is an efficient approach to the problem. The HU aims at decomposing the mixed pixels to obtain the endmembers and abundances, where the endmembers represent the spectra of the pure substances and the abundances are the corresponding percentages of the endmembers in the mixed pixels [8], [9]. The HU models can be roughly divided into the linear mixing model (LMM) and the nonlinear mixing model (NLMM) based on the priors or assumptions used in these models [10]. Compared with the NLMM, the LMM, which regards the observed spectral as a linear combination of some endmembers, has a simple, efficient, and hence it is widely applied in the HU tasks and has been a popular research topic in recent years [11].

In the LMM, the HU can be performed in semi-supervised and unsupervised ways [12], based on whether the spectral library of the endmembers is known or not. In the semi-supervised HU case, where the endmembers of the substances in the scene are known as the large spectral library, the abundances of the mixed pixels are estimated by solving the sparse decomposition problem with some constraints or prior knowledge about the abundances. The SUnSAL [13] and CLSUnSAL [14] use ℓ_1-norm and $\ell_{2,1}$-norm regularizations to constrain the sparsity of the estimated abundances and solve the HU problem with the ADMM algorithm. Besides the sparsity constraints, the spatial information among the pixels of the HSIs is also critical to enhance the performance of the HU. By considering the piecewise smoothness of the HSIs, the SUnSAL-TV [15] uses the total variation (TV) regularization to obtain the more robust result of the HU than the SUnSAL. Moreover, to use more complex spatial information among the pixels, the GLUP-Lap [16], DRSGHU [17], and DSGLSU [18] construct the predefined graph Laplacian from the pixels and introduce the Laplacian regularization, together with the sparse constraints, to solve the HU problem. However, although the semi-supervised unmixing method provides some important insights into unmixing studies, the endmembers in the spectral library have many differences affected by the different illumination, ground, and atmospheric conditions, which inevitably raises some doubts about the HU results [19].

In the unsupervised HU case, the spectral library of the endmembers of the substances in the scene is unknown, and

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hence the HU needs to estimate the endmembers together with the corresponding abundances from the mixed pixels of the HSIs. Deep-learning-based methods [20], [21], [22], such as autoencoders and their variants, have been used for unsupervised HU. In SSABN [21], a spatial–spectral attention bilateral network method for blind unmixing is proposed by combining convolution filters and shifted window transformer block with the autoencoder. In UST-Net [22], a U-shaped transformer network is used to explore the spatial information of the HSI, and then the encoder and decoder processes of the autoencoder are exploited to estimate the abundances and extract the endmembers, respectively. Unfortunately, unsupervised unmixing methods based on autoencoders are sensitive to hyperparameters, and obtaining the optimal solution is difficult [23].

Although many unsupervised HUs based on different strategies or motivations exist in the literature, the nonnegative matrix factorization (NMF)-based HUs [24], [25], [26], [27], [28] have become the prominent branch and have attracted more researchers' attention. To estimate the abundances and the endmembers from the mixed pixels simultaneously, the unsupervised HU needs to incorporate suitable prior knowledge about the abundances and the endmembers [29]. Similar to the semi-supervised HU, the sparsity constraints and spatial information are also crucial for the unsupervised HU to accurately estimate the abundances. In addition to the ℓ_1-norm, the $\ell_{(1/2)}$-norm regularization [30] can further enhance the sparsity of the estimated abundances and improve the performance of the NMF-based HU. Since the abundances can inherit the spatial information of the HSI, the spatial structure regularization is often used to improve the accuracy of the estimated abundances in the HU. In the GLNMF [31], the spatial graph Laplacian is used as the regularization for estimating the abundances in the HU, and it is further extended to the graph TV (gTV) regularization [32]. Moreover, the SSAINMF [33] first transforms the original HSI to the statistical domain to reduce the effects of noise and spectral variability. Then, it uses superpixels to induce local similarity and structural sparsity of abundance within the neighborhood.

Recently, the regularization of the endmembers can be designed for the NMF-based HU based on different strategies or assumptions. The collaborative regularization was introduced in the CoNMF [34] using the reference endmembers obtained from the vertex component analysis (VCA) [35] algorithm and is further extended to the quadratic minimum volume constraint regularization in the QMV [36]. Actually, since distinct endmembers represent different substances, the statistical correlation between them must ideally be very weak [37]. Unfortunately, the spectral signatures of materials are often highly correlated in hyperspectral applications, which is detrimental to the HU algorithm in separating the mixed spectra from the HSI [14]. Therefore, the correlation between endmembers needs to be reduced to estimate a more accurate endmember matrix in HU. With the independence assumption, the KbsNMF [38] and EASNMF [39] based on the NMF framework use different independence constraint strategies for endmember to minimize the correlation of endmembers, respectively.

In compressive sensing (CS), the mutual coherence (MC) is used to measure the degree of correlation between different atoms in the dictionary, and smaller values of MC generally indicate weaker correlation between atoms [40]. The MC is typically minimized during the dictionary learning process to obtain an incoherent dictionary [41]. Since the prior of endmembers in the LMM assumption is similar to the dictionary in CS, minimizing the MC between endmembers in the HU also results in an incoherent endmember matrix. However, it is difficult to directly apply the MC to guide the estimation of the endmember for the HU, since the MC is usually high and tends to be close to one [12]. To solve the above problem, this article proposes a regularization called mutual incoherence (MIC) to constrain the endmembers in HU to promote the incoherence of the estimated endmembers. Moreover, considering the importance of spatial information in HU, a spatial regularization based on relative TV (RTV) is proposed to constrain the abundance and improve the quality of estimated abundance. The main contributions of this article are as follows.

1) As a more general method for defining constraints on endmembers, this article proposes a regularization called MIC to constrain the correlation between endmembers and can improve the quality of estimated endmembers. The proposed MIC regularization can be flexibly implemented with different penalty functions.
2) Instead of directly using the spatial structure of the noisy HSI data, this article proposes a spatial regularization to explore the spatial information of HSI. It can be adaptively learned from the original HSI using graph learning with RTV regularization to extract effective spatial structures and is used to constrain the abundance matrix.
3) With the MIC and RTV regularizations, as well as sparse regularization on abundances, the MRNMF is proposed based on NMF for the HU task using the multiplicative iteration rule. The proposed MRNMF can simultaneously use prior information about the endmember and abundance. The experiments on both synthetic and real HSIs have shown the advantages of the proposed MRNMF over several state-of-the-art approaches.

The rest of this article is organized as follows. Section II briefly introduces the LMM, NMF, and the related NMF-based HUs. In Section III, after the MIC and the RTV regularizations are proposed, the MRNMF is described in detail. The experiments on both the synthetic and real data are present in Section IV. Finally, Section V summarizes the conclusions.

II. Related Work

This section reviews the LMM and the widely applied NMF algorithm in blind HU. Moreover, several NMF-based HUs related to this article are briefly reviewed and analyzed.

A. Linear Mixing Model

The LMM assumes that the mixed pixel of the HSI can be approximated by the linear combination of several endmembers, weighted by their corresponding fractional abundances.

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Given the observed HSI $\mathbf{Y} = [\mathbf{y}_1, \ldots, \mathbf{y}_n] \in \mathbb{R}^{m\times n}$ with m bands and n pixels, the LMM can be formulated as

$$\mathbf{Y} = \mathbf{AX} + \mathbf{N} \tag{1}$$

where $\mathbf{A} = [\mathbf{a}_1, \ldots, \mathbf{a}_p] \in \mathbb{R}^{m\times p}$ denotes the endmember matrix with p spectral signatures, $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_n] \in \mathbb{R}^{p\times n}$ represents the abundance matrix, and $\mathbf{N} \in \mathbb{R}^{m\times n}$ is the additive noise. Usually, the abundance matrix $\mathbf{X}$ is subject to two physical constraints: the abundance nonnegativity constraint (ANC : $\mathbf{X} \geq \mathbf{0}$) and the abundance sum-to-one constraint (ASC : $\mathbf{1}^T\mathbf{X} = \mathbf{1}^T$).

B. Nonnegative Matrix Factorization

The NMF can decompose the nonnegative matrix $\mathbf{Y}$ into the product of two nonnegative matrices. For the unsupervised HU, the NMF-based HU can be formulated as

$$\min_{\mathbf{A},\mathbf{X}} \frac{1}{2}\|\mathbf{Y} - \mathbf{AX}\|_F^2 \quad \text{s.t. } \mathbf{A} \geq \mathbf{0}, \quad \mathbf{X} \geq \mathbf{0} \tag{2}$$

where $\|\cdot\|_F$ denotes the Frobenius norm of the matrix. With the updated scheme [42], the solutions of the NMF-based HU can be iteratively obtained by

$$\mathbf{A} \leftarrow \mathbf{A} \circ \mathbf{YX}^T / \mathbf{AXX}^T \tag{3}$$

$$\mathbf{X} \leftarrow \mathbf{X} \circ \mathbf{A}^T\mathbf{Y} / \mathbf{A}^T\mathbf{AX} \tag{4}$$

where $(\cdot)^T$ is the transpose operation of the matrix, and the "$\circ$" and division are the elementwise operations.

However, the NMF is an ill-posed problem and may result in uncertainty regarding its solution. To overcome this limitation, it is necessary to incorporate some suitable regularizations on the abundance matrix $\mathbf{X}$ and the endmember matrix $\mathbf{A}$, respectively.

C. Related NMF-Based HU Methods

With different regularizations on the abundance matrix $\mathbf{X}$ and the endmember matrix $\mathbf{A}$, several NMF-based HU methods are discussed, including GLNMF [31], gtvMBO [32], CoNMF [34], and EASNMF [39].

1) GLNMF [31]: To use the spatial information in the HSI, the predefined graph Laplacian matrix $\mathbf{L}$ is generated by capturing the local neighborhood relationships using the k-nearest neighbors algorithm on the original pixels of the HSI. By introducing Laplacian regularization and sparse regularization on the abundance matrix $\mathbf{X}$, GLNMF can be formulated as

$$\min_{\mathbf{A},\mathbf{X}} \frac{1}{2}\|\mathbf{Y} - \mathbf{AX}\|_F^2 + \lambda\|\mathbf{X}\|_{\frac{1}{2}} + \frac{\mu}{2}\mathrm{Tr}(\mathbf{XLX}^T) \quad \text{s.t. } \mathbf{A} \geq 0, \quad \mathbf{X} \geq 0, \ \mathbf{1}^T\mathbf{X} = \mathbf{1}^T \tag{5}$$

where $\|\cdot\|_{(1/2)}$ is the $\ell_{(1/2)}$-norm, and λ and μ are the regularization parameters. However, the spatial information captured by $\mathbf{L}$ may be inaccurate and result in poor performance of the HU, since the graph Laplacian $\mathbf{L}$ used in GLNMF is derived directly from distorted or noisy pixels.

2) gtvMBO [32]: Instead of using the graph Laplacian regularization, gtvMBO proposed the gTV regularization $J_{\mathrm{TV}}(\mathbf{X})$ on the abundance matrix $\mathbf{X}$ as follows:

$$J_{\mathrm{TV}}(\mathbf{X}) = \frac{1}{2}\sum_{i,j} \mathbf{W}_{ij}\|\mathbf{x}_i - \mathbf{x}_j\|_1 \tag{6}$$

where $\mathbf{W}_{ij}$ is the edge weight between the pixels $\mathbf{x}_i$ and $\mathbf{x}_j$, and $\|\cdot\|_1$ denotes the ℓ_1-norm. Then, the gtvMBO HU with the gTV regularization is formulated as

$$\min_{\mathbf{A},\mathbf{X}} \frac{1}{2}\|\mathbf{Y} - \mathbf{AX}\|_F^2 + \lambda J_{\mathrm{TV}}(\mathbf{X}) \tag{7}$$

where λ is the regularization parameter. Although the gTV regularization can mitigate the oversmoothing problem caused by the graph Laplacian regularization, the gtvMBO has high computational costs. Unfortunately, the gTV regularization has similar disadvantages to graph Laplacian regularization in the GLNMF, since the weight $\mathbf{W}_{ij}$ in gTV regularization is also derived directly from the distorted or noisy pixels.

3) CoNMF [34]: Using the VCA algorithm, the reference endmember matrix $\mathbf{P}$ is identified from the pixels of the HSI and then is used in the collaborative regularization to constrain the estimated endmember matrix $\mathbf{A}$. With the collaborative regularization, the CoNMF is formulated as

$$\min_{\mathbf{A},\mathbf{X}} \frac{1}{2}\|\mathbf{Y} - \mathbf{AX}\|_F^2 + \alpha\|\mathbf{X}\|_{2,1} + \frac{\beta}{2}\|\mathbf{A} - \mathbf{P}\|_F^2 \tag{8}$$

where $\|\cdot\|_{2,1}$ denotes the $\ell_{2,1}$-norm and $\|\mathbf{X}\|_{2,1} = \sum_{i=1}^{p}\|\mathbf{x}_i\|_2$, and α and β are the regularization parameters. Obviously, the endmember matrix $\mathbf{A}$ estimated in CoNMF is closely related to the reference endmember matrix $\mathbf{P}$. Specifically, the estimated matrix $\mathbf{A}$ needs to locate in the neighborhood of $\mathbf{P}$, controlled by the parameter β. Therefore, the endmember matrix $\mathbf{A}$ is inevitably affected by the accuracy of the reference endmember matrix $\mathbf{P}$.

4) EASNMF [39]: To enhance the accuracy of estimating endmembers in the HU, EASNMF proposes the independent constraint (IC) regularization

$$\Gamma_{\mathrm{IC}}(\mathbf{A}) = \|\mathbf{A}^T\mathbf{A}\|_1 - \|\mathbf{A}\|_F^2 \tag{9}$$

and is formulated as

$$\min_{\mathbf{A},\mathbf{X}} \frac{1}{2}\|\mathbf{Y} - \mathbf{AX}\|_F^2 + \alpha\Gamma_{\mathrm{IC}}(\mathbf{A}) + \beta\|\mathbf{W} \circ \mathbf{X}\|_{\frac{1}{2}} + \frac{\gamma}{2}\mathrm{Tr}\big(\mathbf{XLX}^T\big) \tag{10}$$

where $\|\cdot\|_{(1/2)}$ denotes, and α, β, and λ are the regularization parameters. However, the IC regularization is nonconvex and leads to the local minimum of the HU. In addition, the Laplacian $\mathbf{L}$ in regularization $\mathrm{Tr}(\mathbf{XLX}^T)$ is also predefined based on distorted or noisy images.

III. MIC and RTV Regularizations on NMF

As mentioned above, to handle the ill-posed problem of NMF for unsupervised HU, NMF-based HU needs to impose some proper regularizations on the endmember matrix and the abundance matrix. In this section, the MIC and RTV regularization are proposed to constrain the endmembers and abundances, respectively. By introducing the MIC and RTV

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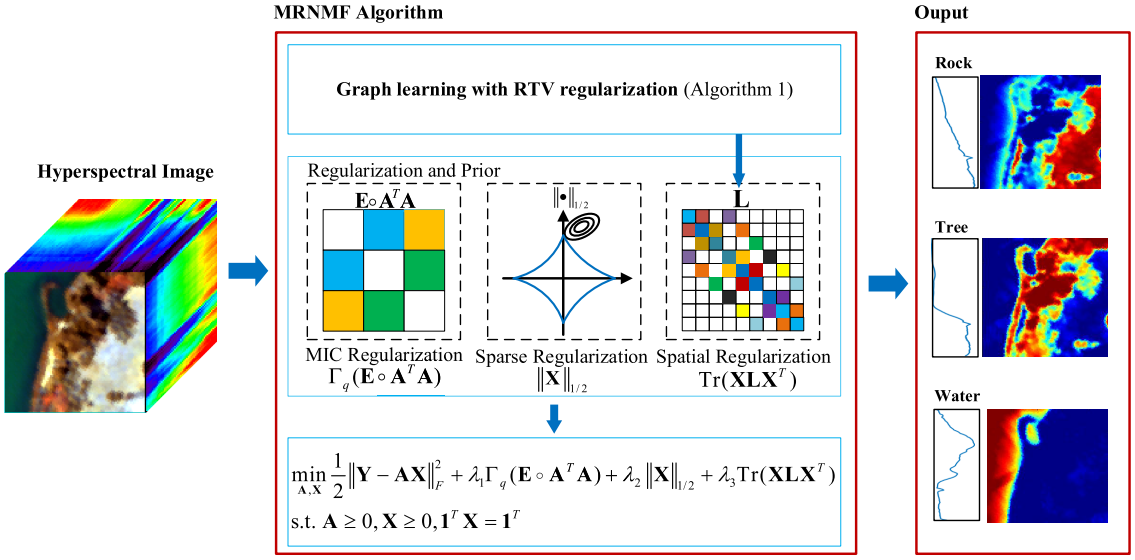


Fig. 1. Flowchart of the MRNMF algorithm for HU.

regularizations, as well as sparse regularization, the MRNMF algorithm is presented using the multiplicative iteration rule. The flowchart of the proposed MRNMF is shown in Fig. 1.

A. MIC Regularization on Endmembers

In the blind HU, the endmembers contain important prior information, and imposing proper constraints on the endmembers can further improve the quality of the estimated endmembers [29], [38]. According to the theories in sparse representation [43], the MC of the endmember matrix $\mathbf{A}$ is defined by

$$\mu(\mathbf{A}) \triangleq \max_{i\neq j} \frac{|\mathbf{a}_i^T\mathbf{a}_j|}{\|\mathbf{a}_i\|^2\|\mathbf{a}_j\|^2} \tag{11}$$

where $\mathbf{a}_i$ and $\mathbf{a}_j$ are the spectral signatures of the endmember matrix $\mathbf{A}$, and $|\cdot|$ is the absolute value function. In sparse representation [43], the quality of the abundances $\mathbf{X}$ depends on the MC of the endmember matrix $\mathbf{A}$. Specifically, the lower the value $\mu(\mathbf{A})$ of the MC, the better the quality of the abundance matrix $\mathbf{X}$. As mentioned previously, it is difficult to apply MC directly to HU tasks. However, recent results [29], [38], [39] show that encouraging endmembers to be independent of each other, instead of minimizing the MC, can help improve the estimation of the endmember matrix $\mathbf{A}$ for the HU.

Base on all the above results, the MIC regularization on the endmember matrix $\mathbf{A}\in\mathbb{R}^{m\times p}$ for the NMF-based HU is proposed as follows:

$$\Gamma(\mathbf{A}) \triangleq \sum_{i\neq j} \psi\left(\mathbf{a}_i^T\mathbf{a}_j\right) \tag{12}$$

where $\psi(\cdot)$ is the sparseness-inducing penalty function, e.g., $|\cdot|$, $(\cdot)^2$, and $\log(|\cdot|+\epsilon)$. For simplicity, the MIC regularization can be written in matrix form as follows:

$$\Gamma_q(\mathbf{A}^T\mathbf{A}) \triangleq \|\mathbf{A}^T\mathbf{A}\circ\mathbf{B}\|_q^q \tag{13}$$

where $\|\cdot\|_q^q$ denotes the ℓ_q-norm for $0<q\leq 2$, and $\mathbf{B}\in\mathbb{R}^{p\times p}$ is the matrix with zero entries on the diagonal and one entries on the off-diagonal, i.e., $\mathbf{B}=\mathbf{1}\mathbf{1}^T-\mathbf{I}$ and $\mathbf{I}$ is the identity matrix. Specifically, the MIC regularization in (13) with $q=1$ and $q=2$ corresponds to the penalty function $\psi(\cdot)$ in (12) with $|\cdot|$ and $(\cdot)^2$, respectively.

The proposed MIC regularization characterizes the overall coherence of the endmember matrix $\mathbf{A}$ by measuring the coherence between each pair of endmembers in $\mathbf{A}$, which has a close relationship with the MC [43] and the IC [39] of endmember. The MC in (11) is consistent with the MIC by replacing the "sum" function with the "max" function and setting $\psi(\cdot)=|\cdot|$ in (12). The significant difference is that the MC measures the maximum similarity among all the endmember pairs, while the MIC characterizes the overall similarity of all the endmember pairs. When the MIC regularization is incorporated into the NMF model, it can encourage all the endmembers to be incoherent with each other. Moreover, by comparing the IC in (9) with the MIC in (13), the former is a particular case of the latter with $q=1$, i.e., $\|\mathbf{A}^T\mathbf{A}\|_1-\|\mathbf{A}\|_F^2=\Gamma_1(\mathbf{A}^T\mathbf{A})$. However, the IC regularization is nonconvex with respect to the endmember matrix $\mathbf{A}$, while the MIC regularization is convex, which can ensure the convergence of the optimization algorithm.

B. Graph Learning With RTV Regularization

As an important prior for the HU, the spatial structure of the HSI can be formulated as spatial regularization on the abundances to improve the performance of NMF-based HU. To characterize the spatial structure, the graph Laplacian is constructed directly from the raw pixels based on the small spatial neighborhood of those pixels. Unfortunately, the raw pixels of the HSI are often contaminated or distorted during the acquisition process due to illumination and atmospheric conditions, as well as noise. Therefore, this graph Laplacian may not accurately capture the spatial structure of the HSI, leading to unreliable results for the HU. To overcome the

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Algorithm 1 Graph Learning With RTV Regularization

Input: noisy image $\mathbf{y}$ and λ_g
1. **Initialize**: $\mathbf{s}^0 = \mathbf{y}$
2. **for** $t = 1$ to K **do**
3. $\quad \mathbf{u}^t = \mathbf{D}\mathbf{s}^t$
4. $\quad$ Calculate $\mathbf{W}(\mathbf{u}^t)$ with (17)
5. $\quad \mathbf{L}(\mathbf{u}^t) = \mathbf{D}\mathbf{W}(\mathbf{u}^t)\mathbf{D}$
6. $\quad$ Update $\mathbf{s}^{t+1}$ with (23)
7. **end for**

Output: graph Laplacian $\mathbf{L} = \mathbf{L}(\mathbf{u}^K)$

Algorithm 2 MRNMF Algorithm

Input: $\mathbf{Y}$, q, and regularization parameters λ_g, λ_1, λ_2, λ_3
1. Learn $\mathbf{L}$ with Algorithm 1
2. **Initialize**: $\mathbf{A}^0$ and $\mathbf{X}^0$
3. **for** $t = 1, 2, \ldots, T$ **do**
4. $\quad$ Update $\mathbf{A}^{t+1}$ with (30)
5. $\quad$ Define $\tilde{\mathbf{A}}^{t+1}$ and $\tilde{\mathbf{Y}}$ with (33)
6. $\quad$ Update $\mathbf{X}^{t+1}$ with (37)
7. **end for**

Output: $\mathbf{A}$ and $\mathbf{X}$

aforementioned problem, we suggest that the graph Laplacian be learned adaptively from the mixed pixels to mitigate the effects of distortion and noise.

In image processing, the RTV regularization [44] can be used to measure the relationship between each pixel and its spatial neighbors, to extract structure information from the texture in an image. Given a clean image $\mathbf{s}$, the RTV regularization is defined in the anisotropic form by

$$\mathrm{RTV}(\mathbf{s}) = \sum_i (\log(|(\mathbf{g}_\sigma * \mathbf{d}_r * \mathbf{s})_i| + \epsilon) + \log(|(\mathbf{g}_\sigma * \mathbf{d}_c * \mathbf{s})_i| + \epsilon)) \quad (14)$$

where " $*$ " denotes the convolution operator, $(\mathbf{z})_i$ represents the i element of the vector $\mathbf{z}$, ϵ is a small positive value, $\mathbf{d}_r$ and $\mathbf{d}_c$ are the first-order difference filters $[-1, 1]$ corresponding to row and column directions, respectively, and $\mathbf{g}_\sigma$ is the Gaussian filter with the scale parameter σ. For simplicity, the RTV regularization in a single direction can be formulated as

$$\mathrm{RTV}(\mathbf{s}) = \sum_i \log(|(\mathbf{G}\mathbf{D}\mathbf{s})_i| + \epsilon) \quad (15)$$

where $\mathbf{G}$ and $\mathbf{D}$ are the convolution matrices corresponding to the convolution filters $\mathbf{g}_\sigma$ and the difference filter $\mathbf{d}$, respectively.

However, the direct use of RTV in the NMF framework is challenging for the algorithm's solution. The following theorem, which is proved in the Appendix, provides a majorizer for the RTV regularization.

Theorem 1: For any vector $\mathbf{u}$ whose length is equal to the number of rows of $\mathbf{D}$, the RTV regularization satisfies the following inequality:

$$\mathrm{RTV}(\mathbf{s}) \le \mathbf{s}^T \mathbf{L}(\mathbf{u})\mathbf{s} + C \quad (16)$$

where "=" holds true for $\mathbf{u} = \mathbf{D}\mathbf{s}$, C is a constant, the graph Laplacian matrix $\mathbf{L}(\mathbf{u}) = \mathbf{D}^T\mathbf{W}(\mathbf{u})\mathbf{D}$, and the diagonal matrix $\mathbf{W}(\mathbf{u})$ is defined by

$$(\mathbf{W}(\mathbf{u}))_{ii} = \left(\mathbf{G}\frac{1}{|(\mathbf{G}\mathbf{u})| + \epsilon} \circ \frac{1}{|\mathbf{u}| + \epsilon}\right)_i. \quad (17)$$

Then, the right term in (16) is the majorizer of $\mathrm{RTV}(\mathbf{s})$.

In image denoising, it is usually assumed that the noisy image $\mathbf{y}$ is the sum of the clean image $\mathbf{s}$ and the disturbing noise $\mathbf{n}$, that is, $\mathbf{y} = \mathbf{s} + \mathbf{n}$. With RTV regularization, the denoised image can be estimated by solving the smoothing filtering problem

$$\min_{\mathbf{s}} \frac{1}{2}\|\mathbf{y} - \mathbf{s}\|_2^2 + \lambda_g \mathrm{RTV}(\mathbf{s}) \quad (18)$$

where $\lambda_g > 0$ is the regularization parameter. With the majorization–minimization (MM) algorithm framework [45], the problem in (18) can be solved by the optimization problem, using a majorizer for the RTV regularization provided in Theorem (1), as follows:

$$\min_{\mathbf{s},\mathbf{u}} \frac{1}{2}\|\mathbf{s} - \mathbf{y}\|_2^2 + \lambda_g \mathbf{s}^T \mathbf{L}(\mathbf{u})\mathbf{s}. \quad (19)$$

This problem can alternatively be solved by the two subproblems of $\mathbf{s}$ and $\mathbf{u}$ in a finite number of iterations.

At the tth iteration, by fixing $\mathbf{s}^t$, $\mathbf{u}$ can be solved by the $\mathbf{u}$-subproblem

$$\mathbf{u}^t = \arg\min_{\mathbf{u}} (\mathbf{s}^t)^T \mathbf{L}(\mathbf{u})\mathbf{s}^t \quad (20)$$

and is updated by

$$\mathbf{u}^t = \mathbf{D}\mathbf{s}^t \quad (21)$$

where $\mathbf{s}^0 = \mathbf{y}$. The above solution is obtained from the fact that $\mathbf{u}^t = \mathbf{D}\mathbf{s}^t$ is the minimum of $(\mathbf{s}^t)^T\mathbf{L}(\mathbf{u})\mathbf{s}^t$ when "=" holds true in (16). By fixing $\mathbf{u}^t$ and the graph Laplacian $\mathbf{L}(\mathbf{u}^t) = \mathbf{D}\mathbf{W}(\mathbf{u}^t)\mathbf{D}$, $\mathbf{s}$ can be obtained by solving the $\mathbf{s}$-subproblem

$$\mathbf{s}^{t+1} = \arg\min_{\mathbf{s}} \frac{1}{2}\|\mathbf{s} - \mathbf{y}\|_2^2 + \lambda_g \mathbf{s}^T \mathbf{L}(\mathbf{u}^t)\mathbf{s} \quad (22)$$

which has the closed-form solution

$$\mathbf{s}^{t+1} = \left(\mathbf{I} + \lambda_g \mathbf{L}(\mathbf{u}^t)\right)^{-1}\mathbf{y}. \quad (23)$$

After K iterations or until convergence, $\mathbf{u}^K$ and $\mathbf{s}^{(K)}$ are obtained, as well the graph Laplacian $\mathbf{L} = \mathbf{L}(\mathbf{u}^K)$ at $\mathbf{u}^K$. With the denoising process, the graph Laplacian $\mathbf{L}$ is adaptively learned from the noisy image $\mathbf{y}$, which captures the spatial structure of $\mathbf{y}$, and the RTV regularization on $\mathbf{s}$ can be represented as $\mathbf{s}^T\mathbf{L}\mathbf{s}$. This process of graph learning is summarized in Algorithm 1.

C. MRNMF Algorithm

Let HSI $\mathbf{Y} \in \mathbb{R}^{m\times n}$ be the distorted or noisy HSI, and $\mathbf{A} \in \mathbb{R}^{m\times p}$ and $\mathbf{X} \in \mathbb{R}^{p\times n}$ are the endmember and abundance matrices. With Algorithm 1, the graph Laplacian $\mathbf{L} \in \mathbb{R}^{n\times n}$ is adaptively learned from the noisy HSI $\mathbf{Y}$. Since the abundance matrix $\mathbf{X}$ is assumed to share the same spatial structures as the HSI $\mathbf{Y}$, the RTV regularization on $\mathbf{X}$ can be formulated as $\mathrm{Tr}(\mathbf{X}\mathbf{L}\mathbf{X}^T)$ in the NMF-based HU.

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Based on MIC regularization on the endmember matrix $\mathbf{A}$ and RTV regularization on abundance matrix $\mathbf{X}$, the NMF-based HU, named MRNMF, is formulated as follows:

$$\begin{aligned}\min_{\mathbf{A},\mathbf{X}} \frac{1}{2}\|\mathbf{Y}-\mathbf{A}\mathbf{X}\|_F^2 + \lambda_1\Gamma_q(\mathbf{E}\circ\mathbf{A}^T\mathbf{A}) + \lambda_2\|\mathbf{X}\|_{\frac{1}{2}} \\ + \lambda_3\mathrm{Tr}(\mathbf{X}\mathbf{L}\mathbf{X}^T) \\ \text{s.t. } \mathbf{A}\geq\mathbf{0}, \quad \mathbf{X}\geq\mathbf{0},\ \mathbf{1}^T\mathbf{X}=\mathbf{1}^T\end{aligned} \tag{24}$$

where $\|\cdot\|_{(1/2)}$ is the $\ell_{(1/2)}$-norm, which induces the sparsity of abundance $\mathbf{X}$ and is defined as $\|\mathbf{X}\|_{(1/2)} = \sum_{i,j}|\mathbf{X}_{ij}|^{(1/2)}$, $\mathbf{E}\in\mathbb{R}^{p\times p}$ is the weighted matrix, and λ_1, λ_2, and λ_3 are the regularization parameters. The weighted matrix $\mathbf{E}$ in $\Gamma_q(\mathbf{E}\circ\mathbf{A}^T\mathbf{A})$ can further promote the MIC property of the endmember matrix $\mathbf{A}$ and can be defined using the iterative reweighted approach. Specifically, at the t-iteration, the weighted matrix $\mathbf{E}$ is defined, using the current endmember matrix $\mathbf{A}^t$, as

$$(\mathbf{E})_{ij} = \frac{1}{|(\mathbf{A}^{tT}\mathbf{A}^t)_{ij}|^q+\epsilon} \tag{25}$$

where ϵ is a small positive value.

The optimization problem of the MRNMF in (24) can be solved with the multiplicative iteration rule [31]. By introducing two dual matrices $\mathbf{\Phi}\in\mathbb{R}^{m\times p}$ and $\mathbf{\Psi}\in\mathbb{R}^{p\times n}$, the Lagrangian is written as

$$\begin{aligned}L(\mathbf{A},\mathbf{X},\mathbf{\Phi},\mathbf{\Psi}) = \frac{1}{2}\|\mathbf{Y}-\mathbf{A}\mathbf{X}\|_F^2 + \lambda_1\Gamma_q(\mathbf{E}\circ\mathbf{A}^T\mathbf{A}) \\ + \lambda_2\|\mathbf{X}\|_{\frac{1}{2}} + \lambda_3\mathrm{Tr}(\mathbf{X}\mathbf{L}\mathbf{X}^T) \\ + \mathrm{Tr}(\mathbf{\Phi}^T\mathbf{A}) + \mathrm{Tr}(\mathbf{\Psi}^T\mathbf{X}) \\ \text{s.t. } \mathbf{1}^T\mathbf{X}=\mathbf{1}^T, \quad \mathbf{\Phi}\geq\mathbf{0},\ \mathbf{\Psi}\geq\mathbf{0}.\end{aligned} \tag{26}$$

The minimum of the Lagrangian function in (26) can be achieved by alternately minimizing two subproblems: the $\mathbf{A}$-subproblem and the $\mathbf{X}$-subproblem, with respect to $\mathbf{A}$ and $\mathbf{X}$, respectively.

The $\mathbf{A}$-subproblem, by fixing $\mathbf{X}$, can be formulated as

$$\min_{\mathbf{A}} \frac{1}{2}\|\mathbf{Y}-\mathbf{A}\mathbf{X}\|_F^2 + \lambda_1\Gamma_q(\mathbf{E}\circ\mathbf{A}^T\mathbf{A}) + \mathrm{Tr}(\mathbf{\Phi}^T\mathbf{A}). \tag{27}$$

The Karush–Kuhn–Tucker (KKT) conditions of the above subproblem are

$$\mathbf{0} = \mathbf{A}\mathbf{X}\mathbf{X}^T - \mathbf{Y}\mathbf{X}^T + \lambda_1\nabla_{\mathbf{A}}\Gamma_q(\mathbf{E}\circ\mathbf{A}^T\mathbf{A}) + \mathbf{\Phi} \tag{28}$$

$$\mathbf{0} = \mathbf{\Phi}\circ\mathbf{A}. \tag{29}$$

Then, the update rule for $\mathbf{A}$ is

$$\mathbf{A}^{t+1} = \mathbf{A}^t \circ \frac{\mathbf{Y}\mathbf{X}^{tT}}{\mathbf{A}^t\mathbf{X}^t\mathbf{X}^{tT} + \lambda_1\nabla_{\mathbf{A}}\Gamma_q(\mathbf{E}\circ\mathbf{A}^{tT}\mathbf{A}^t)} \tag{30}$$

where the operators "$\circ$" and "$(\cdot/\cdot)$" denote elementwise multiplication and elementwise division, respectively. The gradient of $\Gamma_q(\mathbf{E}\circ\mathbf{A}^T\mathbf{A})$ with respect to variable $\mathbf{A}$, for different values of q, is given by

$$\nabla_{\mathbf{A}}\Gamma_q(\mathbf{E}\circ\mathbf{A}^{tT}\mathbf{A}^t) = \begin{cases} 2\mathbf{A}^t\mathbf{H}, & \text{for } q=1 \\ 4\mathbf{A}^t\left((\mathbf{A}^{tT}\mathbf{A}^t)\circ\mathbf{H}^{\cdot 2}\right), & \text{for } q=2 \end{cases} \tag{31}$$

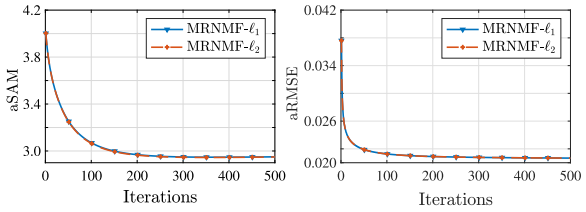


Fig. 2. Convergence curves of the proposed method on the synthetic dataset.

where $\mathbf{H}=\mathbf{E}\circ\mathbf{B}$, the weighted matrix $\mathbf{E}$ is defined in (25), the matrix $\mathbf{B}=\mathbf{1}\mathbf{1}^T-\mathbf{I}$, and $\mathbf{H}^{\cdot 2}$ denotes the square of the elements of the matrix $\mathbf{H}$.

Next, the $\mathbf{X}$-subproblem, with fixed $\mathbf{A}$, can be written as

$$\begin{aligned}\min_{\mathbf{X}} \frac{1}{2}\|\mathbf{Y}-\mathbf{A}\mathbf{X}\|_F^2 + \lambda_2\|\mathbf{X}\|_{\frac{1}{2}} + \lambda_3\mathrm{Tr}(\mathbf{X}\mathbf{L}\mathbf{X}^T) \\ + \mathrm{Tr}(\mathbf{\Psi}^T\mathbf{X}) \\ \text{s.t. } \mathbf{1}^T\mathbf{X}=\mathbf{1}^T.\end{aligned} \tag{32}$$

To eliminate the equality constraint, define

$$\tilde{\mathbf{A}} = \begin{bmatrix}\mathbf{A} \\ \delta\mathbf{1}^T\end{bmatrix}, \quad \tilde{\mathbf{Y}} = \begin{bmatrix}\mathbf{Y} \\ \delta\mathbf{1}^T\end{bmatrix} \tag{33}$$

where $\delta>0$. The $\mathbf{X}$-subproblem can be equivalently written as

$$\min_{\mathbf{X}} \frac{1}{2}\|\tilde{\mathbf{Y}}-\tilde{\mathbf{A}}\mathbf{X}\|_F^2 + \lambda_2\|\mathbf{X}\|_{\frac{1}{2}} + \lambda_3\mathrm{Tr}(\mathbf{X}\mathbf{L}\mathbf{X}^T) + \mathrm{Tr}(\mathbf{\Psi}^T\mathbf{X}). \tag{34}$$

The KKT conditions of the $\mathbf{X}$-subproblem are

$$\mathbf{0} = \tilde{\mathbf{A}}^T\tilde{\mathbf{A}}\mathbf{X} - \tilde{\mathbf{A}}^T\tilde{\mathbf{Y}} + \frac{\lambda_2}{2}\mathbf{X} + 2\lambda_3\mathbf{X}\mathbf{L} + \mathbf{\Psi} \tag{35}$$

$$\mathbf{0} = \mathbf{\Psi}\circ\mathbf{A}. \tag{36}$$

Therefore, at the $(t+1)$-iteration, $\mathbf{X}$ can be updated by

$$\mathbf{X}^{t+1} = \mathbf{X}^t \circ \frac{\tilde{\mathbf{A}}^{(t+1)T}\tilde{\mathbf{Y}} + 2\lambda_3\mathbf{X}^t\mathbf{C}}{\tilde{\mathbf{A}}^{(t+1)T}\tilde{\mathbf{A}}^{t+1}\mathbf{X}^t + \frac{\lambda_2}{2}(\mathbf{X}^t)^{-\frac{1}{2}} + 2\lambda_3\mathbf{X}^t\mathbf{Z}} \tag{37}$$

where $\mathbf{Z}=\mathrm{Diag}(\mathbf{L})$ and $\mathbf{C}=\mathbf{Z}-\mathbf{L}$.

With the updated rules in (30) and (37), the iterative procedure of the MRNMF is summarized in Algorithm 2. The MRNMF algorithm is guaranteed to converge to a local minimum of the optimization problem (24) and satisfies the KKT optimality conditions in (28), (29), (35), and (36). Moreover, Fig. 2 demonstrates the convergence results of the MRNMF algorithm, showing that the MRNMF reaches a stable solution after 300 iterations.

IV. Experiments

In this section, several experiments are conducted to evaluate and demonstrate the performance of the proposed MRNMF algorithm, compared with four state-of-the-art NMF-based blind HU methods, including GLNMF [31], CoNMF [34], gtvMBO [32], and EASNMF [39], as well as a deep-learning-based SSABN [21] method. With a different value of q in the MIC regularization (13), MRNMF can be implemented as MRNMF-ℓ_1 for $q=1$ and MRNMF-ℓ_2 for $q=$. In these experiments, the performances of these HU methods are

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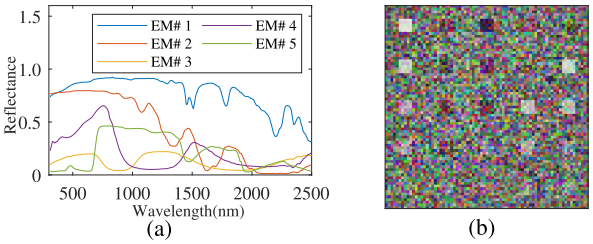


Fig. 3. Five endmembers and false-color image of the synthetic dataset with SNR = 10 dB. (a) Endmember. (b) False-color image.

evaluated using four indices [32], [46]: average spectral angle mapper (aSAM), endmember overall root mean square error (eRMSE), abundance overall root mean square error (aRMSE), and reconstruction overall root mean square error (rRMSE). The aSAM is defined as

$$\text{aSAM}(\mathbf{A}, \hat{\mathbf{A}}) = \frac{1}{p}\sum_{j=1}^{p} \arccos\left(\frac{\mathbf{a}_j^T \hat{\mathbf{a}}_j}{\|\mathbf{a}_j\|_2 \|\hat{\mathbf{a}}_j\|_2}\right)$$

and the eRMSE is defined as

$$\text{eRMSE}(\mathbf{A}, \hat{\mathbf{A}}) = \frac{1}{p}\sum_{j=1}^{p} \|\mathbf{a}_j - \hat{\mathbf{a}}_j\|_2$$

where $\mathbf{A} \in \mathbb{R}^{m\times p}$ and $\hat{\mathbf{A}} \in \mathbb{R}^{m\times p}$ represent the reference and estimated endmember matrices, respectively. When the reference abundances are available, the aRMSE is defined by

$$\text{aRMSE}(\mathbf{X}, \hat{\mathbf{X}}) = \frac{1}{n}\sum_{i=1}^{n} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2$$

where $\mathbf{X} \in \mathbb{R}^{p\times n}$ and $\hat{\mathbf{X}} \in \mathbb{R}^{p\times n}$ are the reference and estimated abundance matrices, respectively, and $\mathbf{x}_i$ and $\hat{\mathbf{x}}_i$ are the ith column vector of $\mathbf{X}$ and $\hat{\mathbf{X}}$, respectively. Without reference abundances, the rRMSE is defined by

$$\text{rRMSE}(\mathbf{Y}, \hat{\mathbf{Y}}) = \frac{1}{n}\sum_{i=1}^{n} \|\mathbf{y}_i - \hat{\mathbf{y}}_i\|_2$$

where $\hat{\mathbf{Y}} \in \mathbb{R}^{m\times n}$ is the reconstruction of the HSI data $\mathbf{Y}$.

A. Experimental Setup

To implement the proposed MRNMF and four compared HU algorithms, the endmember matrix $\mathbf{A}$ and abundance matrics $\mathbf{X}$ need to be initialized first. For a fair comparison, all these algorithms use the same way to initialize $\mathbf{A}$ and $\mathbf{X}$ in all the numerical experiments. The number of endmembers, denoted by p, is an important parameter in the blind HU task. In the experiments, the value of p for MRNMF is known a priori and is estimated by endmember extraction algorithms, such as virtual dimensionality (VD) [47] and Hysime [48]. Given the HSI data $\mathbf{Y}$, the initialized endmember matrix $\mathbf{A}^0$ is obtained using the VCA algorithm [35] with the same number of the endmembers. With the initialized $\mathbf{A}^0$, the initialized abundance matrix $\mathbf{X}^0$ is obtained using the SUnSAL algorithm [13]. In MRNMF, the ANC for $\mathbf{A}^t$ and $\mathbf{X}^t$ in Algorithm (2) is naturally satisfied based on the multiplicative iteration rule [7], since $\mathbf{A}^0$ and $\mathbf{X}^0$ are nonnegative. The ASC in the $\mathbf{X}$-subproblem can be equivalently written as an unconstrained optimization problem using (33), where the parameter δ is set to 15. Moreover, the stopping criterion of the proposed MRNMF algorithm relies on the error tolerance tol and the maximum number T of iterations. In these experiments, tol and T are set to 10^{-3} and 200, respectively, for all the datasets in practice.

B. Experiments on Synthetic Dataset

1) Dataset Description: The synthetic dataset is generated using five reference endmembers with 224 bands randomly selected from USGS [49], as shown in Fig. 3(a). It contains 75×75 pixels arranged in a grid of 25 squares [50], [51], [52], strictly complying with the ANC and ASC. Then, the data cube is contaminated by Gaussian noise with different signal-to-noise ratios (SNRs). Specifically, the reference endmembers of synthetic data and the false-color image with SNR = 10 dB are shown in Fig. 3(a) and (b).

2) Parameters Selection: For a fair comparison, the optimal parameters are selected for all the algorithms to ensure maximum performance. In GLNMF, the parameters λ is set to 1 and μ is set to 5. CoNMF has the parameters $\alpha = 7.5e-4$ and $\beta = 5.61e+1$. The gtvMBO's parameters are set to $\lambda = 1e{-}3.75$, $\rho = 1e{-}1.75$, and $\gamma = 1e+4$. The SSABN's learning rate is set to $4e-3$ and the epoch is set to $2e+2$, while EASNMF has the parameters $\alpha = 1e-3$, $\beta = 1e-1$, and $\gamma = 5e-2$. Moreover, Fig. 4 shows the effect on the performance of MRNMF when only one of its multiple parameters is varied.

For MRNMF-ℓ_1, it can be observed that the value of aRMSE gradually decreases as the parameter λ_1 is increased from $5e-2$ to $1e+1$. However, the value of aSAM increases when λ_1 is greater than $5e-1$. To obtain a comprehensive result, the parameter λ_1 is set to $1e-1$. As λ_2 varies between $5e-1$ and 5, both the aRMSE and aSAM gradually decrease. However, the changes in aSAM values are not obvious when λ_2 is greater than 4, so the final setting is $\lambda_2 = 5$. With an increase in λ_3 from $5e-2$ to $5e-1$, the aRMSE value increases and then stabilizes, while the aSAM value gradually decreases and then stabilizes. Based on this, set λ_3 to $3e-1$. Both the aRMSE and aSAM are minimized at the point when the value for λ_g changes from $1e-3$ to $5e-2$. Therefore, we should set $\lambda_g = 5e-3$. For MRNMF-ℓ_2, as the value of λ_1 increases, the aRMSE converges to a stable value, while the aSAM decreases slightly, eventually setting $\lambda_1 = 1e+1$. The impact of λ_2, λ_3, and λ_g on the aRMSE and aSAM values is similar to the results of MRNMF-ℓ_1.

3) Results and Analysis: Table I lists the quantitative results and computational times of these algorithms on the synthetic dataset with SNR = 10 dB, and Fig. 5 provides the visual comparison. Due to the absence of constraints on the endmember matrix $\mathbf{A}$, GLNMF and SSABN exhibit poor performance, as indicated by the values of aSAM and aRMSE. Although the constraint, based on the estimated endmembers from the VCA algorithm, is introduced into CoNMF, it also produces lower accuracies because spatial information is not used, and the estimation results of VCA are not robust. Compared with GLNMF, SSABN, and CoNMF, the EASNMF

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TABLE I
QUANTITATIVE RESULTS AND COMPUTATIONAL TIMES FOR THE SYNTHETIC DATASET

Methods		GLNMF	CoNMF	gtvMBO	EASNMF	SSABN	MRNMF-ℓ_1	MRNMF-ℓ_2
	Average	6.6997	6.3548	4.1808	3.9519	9.3277	**2.9683**	**2.9643**
	EM# 1	2.2733	2.1996	1.3796	1.4194	2.8171	**1.1745**	**1.1732**
	EM# 2	3.6141	5.0984	2.3498	2.0137	2.3763	**1.6003**	**1.5963**
aSAM	EM# 3	12.3485	13.8417	8.6486	9.3392	30.5529	**6.2837**	**6.2717**
	EM# 4	5.3410	6.8519	4.5189	4.1989	7.1640	**3.1837**	**3.1824**
	EM# 5	9.9213	3.7823	4.0071	2.7880	3.7284	**2.5993**	**2.5978**
aRMSE		0.0646	0.0468	0.0427	0.0510	0.0991	**0.0209**	**0.0209**
eRMSE		0.0491	0.0529	0.0357	0.0388	0.2173	**0.0278**	**0.0278**
aRMSE+eRMSE		0.1137	0.0997	0.0784	0.0898	0.3164	**0.0487**	**0.0487**
Graph Time (s)		17.34	-	**0.02**	13.72	-	0.10	0.14
Algotirhm Time (s)		18.80	**0.24**	1.00	24.74	120.34	0.94	1.02

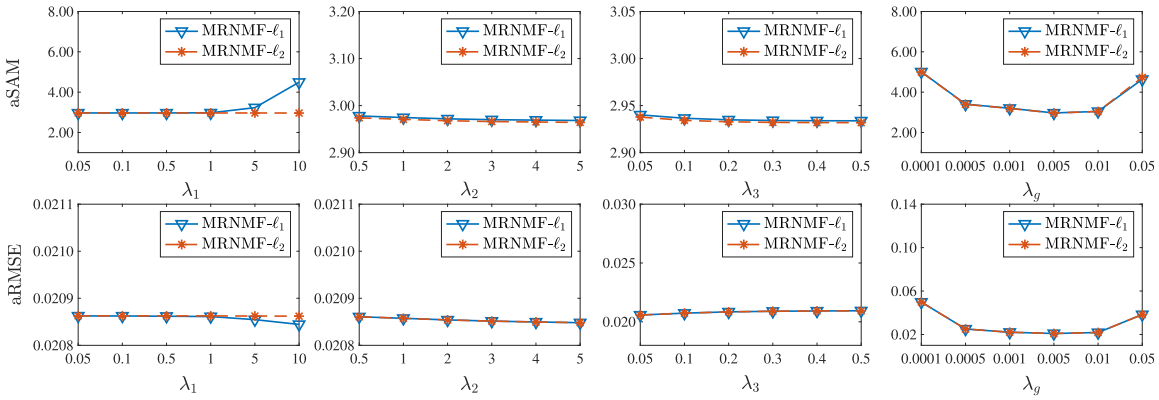


Fig. 4. Performance of MRNMF on the synthetic dataset with respect to parameters λ_1, λ_2, λ_3, and λ_g.

TABLE II
QUANTITATIVE RESULTS AND COMPUTATIONAL TIMES FOR THE SAMSON DATASET

Methods		GLNMF	CoNMF	gtvMBO	EASNMF	SSABN	MRNMF-ℓ_1	MRNMF-ℓ_2
	Average	3.9048	4.6639	8.3534	2.6804	4.3882	**2.0829**	**2.3133**
	Rock	**1.2801**	1.4462	2.6349	1.4314	7.7578	1.6445	1.6489
aSAM	Tree	3.5442	3.0128	5.4478	3.6693	2.5495	**1.8900**	**1.8807**
	Water	6.8901	9.5327	23.2506	2.9403	2.8571	**2.7142**	3.4103
aRMSE		0.1360	0.2774	0.1266	0.1273	0.1167	**0.1066**	**0.1066**
eRMSE		0.0716	0.0715	0.0725	0.0744	0.3129	**0.0653**	**0.0653**
aRMSE+eRMSE		0.2075	0.3489	0.1991	0.2017	0.4296	**0.1718**	**0.1718**
Graph Time (s)		30.84	-	**0.03**	22.26	-	0.22	0.23
Algorithm Time (s)		32.55	1.18	**0.66**	40.64	90.67	0.79	0.80

performs better due to IC regularization on the endmember matrix $\mathbf{A}$ and spatial structure regularization on the abundance matrix $\mathbf{X}$. However, gtvMBO still achieves similar performance to EASNMF, possibly due to the effectiveness of gTV regularization on the abundance matrix $\mathbf{X}$. Among all these algorithms, the proposed MRNMF-ℓ_1 and MRNMF-ℓ_2 achieve the highest accuracies, as indicated by the values of aSAM and aRMSE in Table I. In addition, the abundances estimated by the MRNMF are more robust against noise, as shown in Fig. 5. Thus, the performance of MRNMF indicates the effectiveness of MIC and RTV regularizations.

C. Experiments on Samson Dataset

1) Dataset Description: The Samson dataset consists of 95×95 pixels with 156 bands and contains three endmembers [32]. Its wavelength range is from 401 to 889 nm with a spectral resolution of 3.13 nm and has been widely used for HU tasks [51]. Moreover, the reference endmembers and the false-color image of the Samson are shown in Fig. 6.

2) Parameters Selection: The optimal parameters for the HU algorithms for the Samson dataset are selected based on the performance measured by the values of aSAM and aRMSE. For GLNMF, the parameters are set to $\lambda = 5e-1$ and $\mu = 8e-1$. The parameters of CoNMF are set to $\alpha = 9.5e-5$ and $\beta = 2.71e+3$. In gtvMBO, the parameters are set to $\lambda = 1e-3.75$, $\rho = 1e-2.25$, and $\gamma = 1e+4$. The parameters of EASNMF are set to $\alpha = 1e-2$, $\beta = 5e-1$, and $\gamma = 1e-2$. In SSABN, the learning rate and epoch are set to $2e-3$ and $5e+1$, respectively. For the proposed MRNMF methods, the value of λ_1 is set to

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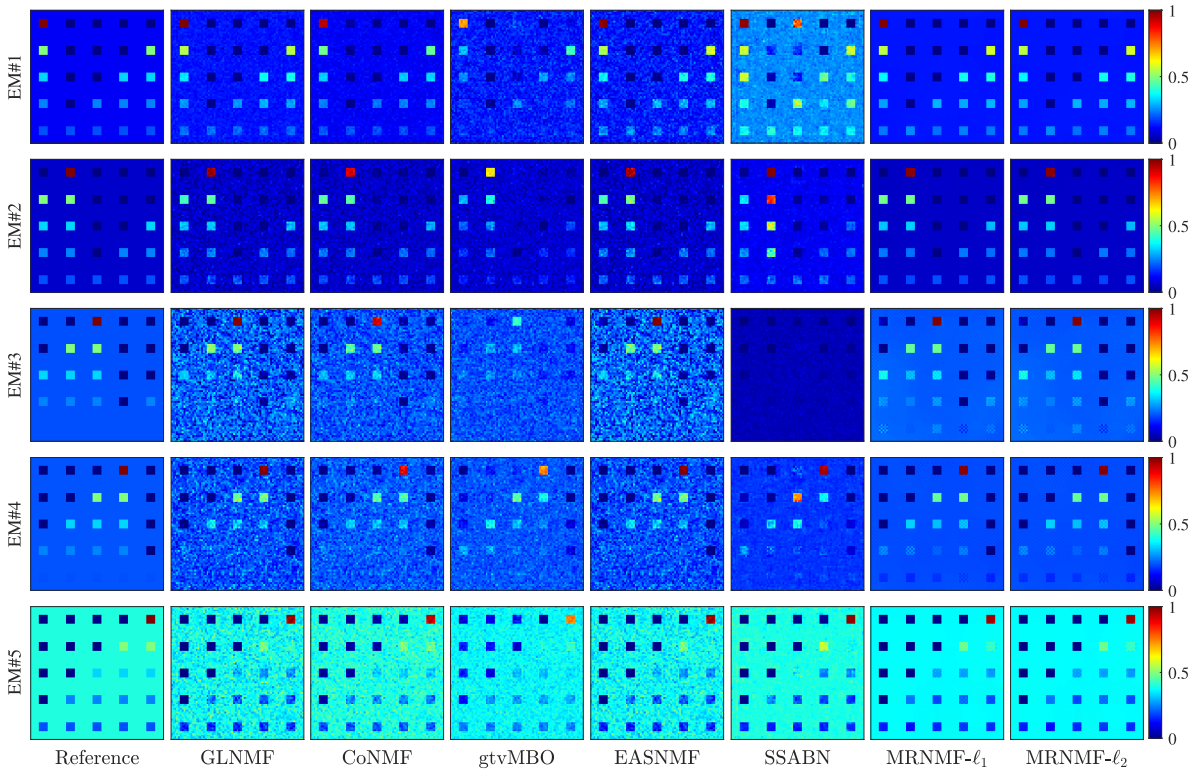


Fig. 5. Estimated abundance maps from the synthetic dataset with SNR 10 dB.

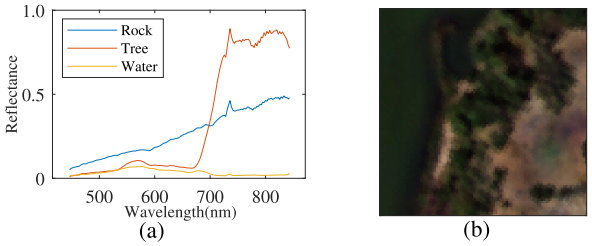


Fig. 6. Reference endmembers and false-color image of the Samson dataset. (a) Endmember. (b) False-color image.

9 and 1.85 for MRNMF-ℓ_1 and MRNMF-ℓ_2, respectively, and other parameters are set to $\lambda_2 = 4$, $\lambda_3 = 6e-2$, and $\lambda_g = 1e-4$.

3) Comparison Results: Table II lists the quantitative results and computational times of these algorithms on the Samson dataset, and the estimated abundance maps are shown in Fig. 7, where the reference abundances are obtained from [53]. As indicated by the values of aSAM and aRMSE, GLNMF and CoNMF achieve poor performance due to the absence or incorrect regularization on the endmembers and abundances. Although gtvMBO achieves the highest accuracy of aRMSE for abundances, due to the effectiveness of gTV regularization on the abundances, it obtains the lowest accuracy of aSAM for endmembers. Compared with gtvMBO, the SSABN incorporates transformer and convolutional filter constraints on the abundance matrix $\mathbf{X}$ and estimates relatively good endmembers and abundances. In EASNMF, IC regularization and spatial structure regularization are used to constrain the endmembers and abundances, respectively, and hence EASNMF yields better performance. Since the effectiveness of MIC and RTV regularizations, the proposed MRNMF-ℓ_1 and MRNMF-ℓ_2 algorithms outperform these competitors in terms of the accuracies of endmembers and abundances.

D. Experiments on Urban Dataset

1) Dataset Description: The Urban dataset is obtained from the HYDICE HSI taken from the urban scene of Copperas Cove, TX, US, and contains four endmembers [32], [46], [54]. The data cube consists of 307 × 307 pixels and retains 162 spectral bands for these pixels after removing dense water vapor and atmospheric effects. The spectral range is from 400 to 2500 nm, and the resolution is 10 nm. The reference endmembers and the false-color image of the Urban are shown in Fig. 8.

2) Parameters Selection: The optimal parameters for all the unmixing methods are selected as follows. The GLNMF parameters λ and μ are set to $1e-1.5$ and $1e-5.5$, respectively. In CoNMF, the parameters are set to $\alpha = 3.07$ and $\beta = 3.77e+4$. For gtvMBO, the parameters are $\lambda = 1e-6$, $\rho = 1e{-}5.5$, and $\gamma = 1e{+}4.25$. The parameters for EASNMF are $\alpha = 1e-2$, $\beta = 5e-1$, and $\gamma = 5e-1$. In SSABN, the learning rate and epoch are set to $6e{-}3$ and $5e{+}1$, respectively. For the MIC regularization $\Gamma_q(\cdot)$, we set the parameters λ_1 to $1e-2$ and $5e-2$ for $q = 1$ and $q = 2$, respectively, and set the other parameters to $\lambda_2 = 8e-1$, $\lambda_3 = 1e-1$, and $\lambda_g = 1e-2$.

3) Comparison Results: Table III lists the quantitative results and computational times of these algorithms on the Urban dataset, and the estimated abundance maps are shown

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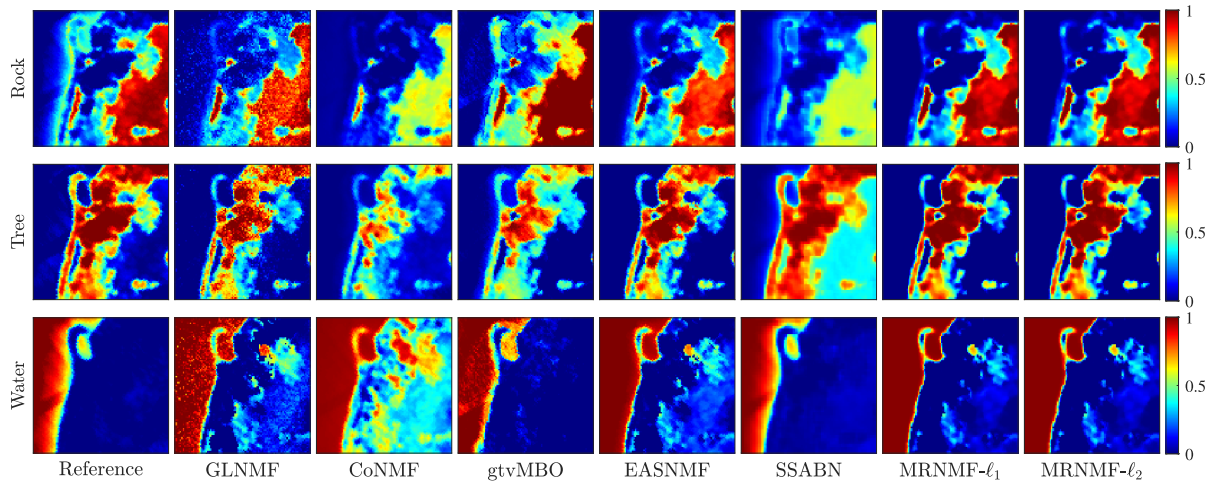


Fig. 7. Abundance maps by different unmixing methods and the reference in the Samson dataset.

TABLE III
QUANTITATIVE RESULTS AND COMPUTATIONAL TIMES FOR THE URBAN DATASET

Methods		GLNMF	CoNMF	gtvMBO	EASNMF	SSABN	MRNMF-ℓ_1	MRNMF-ℓ_2
aSAM	**Average**	18.1093	33.3672	20.9197	20.5606	**11.3331**	16.5315	16.5334
	Asphalt	17.3237	12.2804	11.6760	19.8052	18.0645	**4.9868**	**4.9869**
	Grass	7.6687	99.4410	11.0045	12.6603	14.9927	**3.9984**	**4.0000**
	Tree	12.4200	11.9962	15.5392	5.8224	**3.3269**	8.3119	8.3130
	Roof	35.0249	9.7514	45.4592	43.9545	**8.9481**	48.8291	48.8336
aRMSE		0.2869	0.2800	0.1724	0.2673	0.1953	**0.1633**	**0.1633**
eRMSE		0.2313	0.3248	**0.1301**	0.1750	0.4141	0.1335	0.1335
aRMSE+eRMSE		0.5183	0.6048	0.3025	0.4423	0.6094	**0.2968**	**0.2968**
Graph Time (s)		659.71	-	4.22	621.10	-	3.00	**2.77**
Algorithm Time (s)		689.82	30.77	30.91	981.81	1070.72	7.69	**7.36**

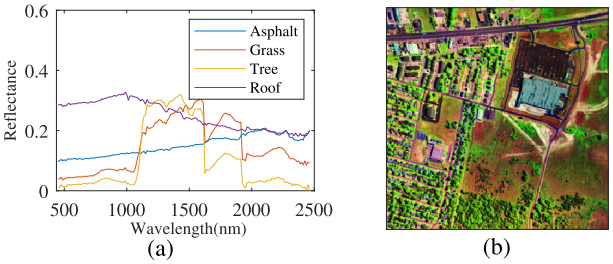


Fig. 8. Reference endmembers and false-color image of the Urban dataset. (a) Endmember. (b) False-color image.

in Fig. 9, where the reference abundances are also obtained from [53]. For the Urban dataset, GLNMF still has poor aRMSE values because the spatial information is extracted based on noisy HSI. In addition, CoNMF and EASNMF yield poor accuracies of aSAM for endmembers and aRMSE for abundances. This may be due to the incorrect regularization of the endmembers, where the estimation result of VCA used in CoNMF is not robust and the IC regularization in EASNMF is ineffective. Both the gtvMBO and SSABN achieve better performance due to the effective regularization of abundance. Moreover, the gtvMBO algorithm, while obtaining good eRMSE values, has poor aSAM values. Similarly, SSABN achieves high aSAM values, but the overall RMSE values for endmembers and abundance are undesirable. Compared with these algorithms, the proposed MRNMF-ℓ_1 and MRNMF-ℓ_2 methods achieve the best performance with the proposed MIC and RTV regularizations.

E. Experiments on Cuprite Dataset

1) Dataset Description: The Cuprite dataset is acquired by the AVIRIS sensor over the Cuprite mine in western Nevada, US, and it is the most typical and challenging dataset in HU [51], [55], [56]. The data cube contains 12 endmembers, cropped from the original HSI image, consists of 250 $\times$ 191 pixels, and retains 188 spectral bands by removing water absorption and low SNR bands. The spectral range is from 400 to 2500 nm, and the resolution is 10 nm. The reference endmembers and the false-color image of the Cuprite are shown in Fig. 10.

2) Parameters Selection: In the Cuprite experiments, the parameters are selected based on aSAM values and empirically observed abundance maps. The GLNMF is set with parameters $\lambda = 1e{-}3.5$ and $\mu = 1e{-}1.5$. CoNMF sets the parameters $\alpha = 2.19e - 6$ and $\beta = 5.73e + 4$. The gtvMBO's parameters are set to $\lambda = 1e - 6$, $\rho = 1e - 4.5$, and $\gamma = 1e + 4.25$. For EASNMF, the parameters are set to $\alpha = 1e - 1$, $\beta = 1e - 2$, and $\gamma = 1e - 1$. In SSABN, the parameters; learning rate and epoch are set to $4e - 3$ and $2e + 2$, respectively. The parameters of MRNMF are set as follows: $\lambda_1 = 1e - 2$ and $1e + 2$ correspond to $q = 1$ and $q = 2$ in regularization

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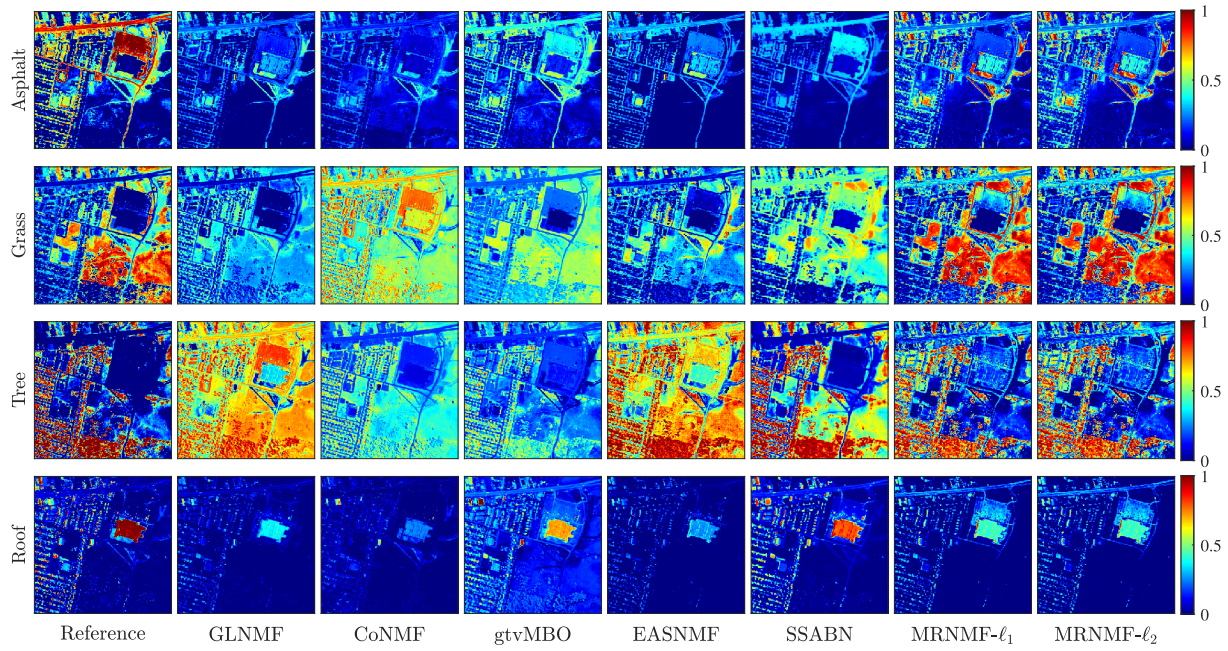


Fig. 9. Abundance maps by different unmixing methods and the reference in the Urban dataset.

TABLE IV
QUANTITATIVE RESULTS AND COMPUTATIONAL TIMES FOR THE CUPRITE DATASET

Methods		GLNMF	CoNMF	gtvMBO	EASNMF	SSABN	MRNMF-ℓ_1	MRNMF-ℓ_2
aSAM	**Average**	4.7769	5.5310	4.7650	4.7290	7.0486	**4.6974**	**4.6972**
	Alunite	6.6471	**4.9732**	6.3946	6.1927	6.1194	6.0462	6.0464
	Andradite	**4.7028**	5.5412	5.3332	4.8044	5.3526	4.8153	4.8154
	Buddingtonite	5.5195	**4.6766**	5.7412	5.3311	11.6580	5.2489	5.2473
	Dumortierite	**4.4062**	4.5036	4.4608	4.5368	5.1802	4.5884	4.5885
	Kaolinite1	4.6691	**4.5372**	5.5671	4.6337	4.6497	4.5912	4.5912
	Kaolinite2	3.9416	4.0935	**3.3675**	4.0990	4.4825	4.1416	4.1418
	Muscovite	8.1931	**7.5387**	8.0730	8.2343	22.8916	8.2440	8.2441
	Montmorillonite	3.2929	3.4798	**3.2888**	3.3025	3.7069	3.3081	3.3080
	Nontronite	4.2511	4.5014	**4.0492**	4.2583	5.0408	4.2648	4.2650
	Pyrope	3.7739	13.9448	**3.1165**	3.7583	4.1218	3.7175	3.7163
	Sphene	**3.4778**	3.6682	3.9882	3.5108	3.5397	3.5100	3.5097
	Chalcedony	**3.4473**	4.9141	3.7995	4.0866	7.8403	3.8933	3.8933
eRMSE		0.2162	**0.1530**	0.2037	0.2132	0.2385	0.2115	0.2116
rRMSE		0.0077	**0.0050**	0.0245	0.0128	0.4391	0.0159	0.0159
Graph Time (s)		181.45	-	1.220	194.83	-	**0.51**	0.55
Algorithm Time (s)		200.09	23.37	12.88	403.83	1181.90	1.81	**1.77**

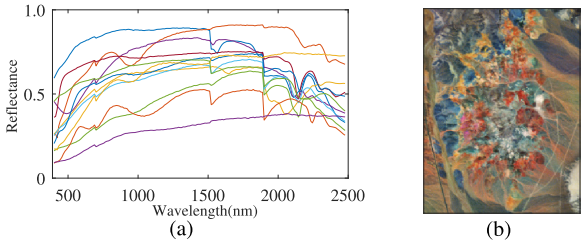


Fig. 10. Reference endmembers and false-color image of the Cuprite dataset. (a) Endmember. (b) False-color image.

$\Gamma_q(\cdot)$, respectively, the parameter λ_2 is set to $1e-1$, and the parameters λ_3 and λ_g are set to $1e-2$.

3) Comparison Results: Table IV lists the quantitative results and computational times of these algorithms on the Cuprite dataset. Since the ground-truth abundances are not available in the Cuprite dataset, the abundances of several representative minerals in the scene, such as Alunite, Buddingtonite, Kaolinite, and Chalcedony, are selected to measure the performance of abundance estimation based on rRMSE. In terms of the average values of aSAM, the proposed MRNMF-ℓ_1 and MRNMF-ℓ_2 methods outperform the other algorithms and exhibit the lowest computational cost. Moreover, Fig. 11 presents a brief visual analysis of the estimated abundance map. It shows that the proposed MRNMF methods are more robust against noise.

F. Robustness Analysis

To investigate the robustness of the HU algorithms, the proposed methods are performed on the synthetic dataset

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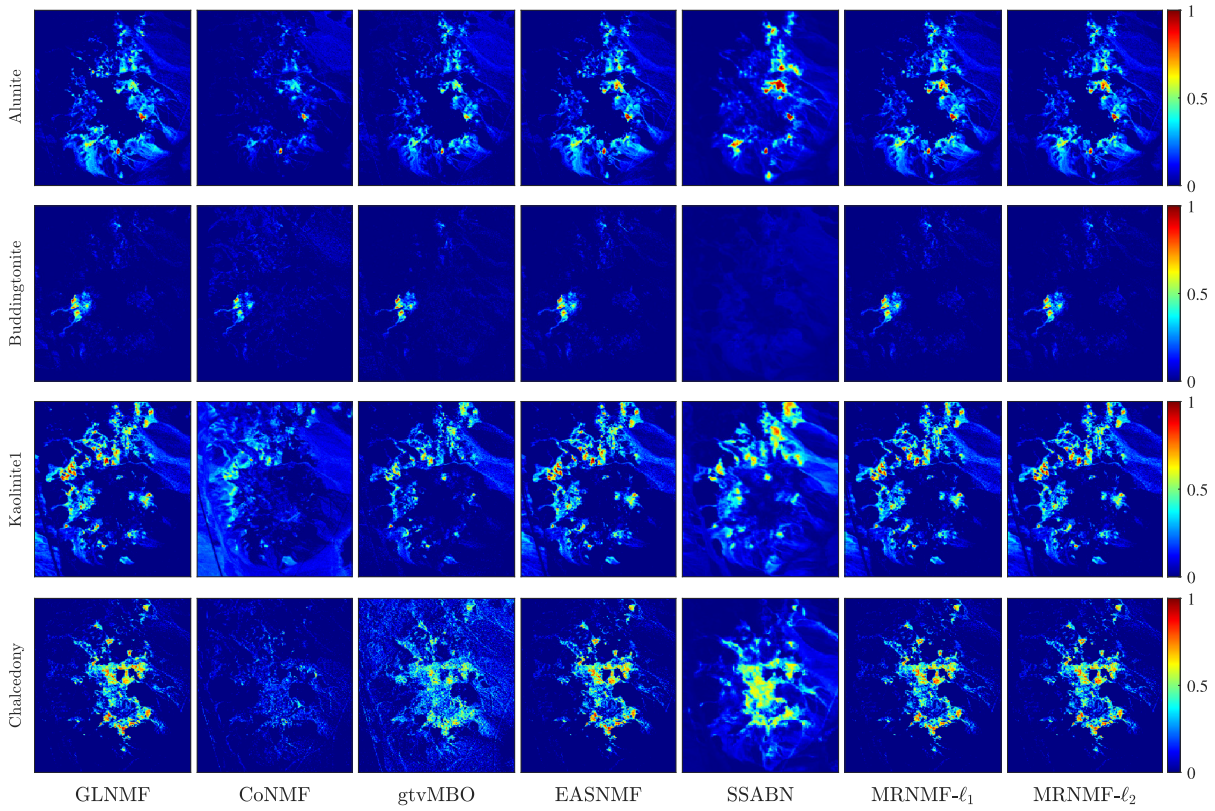


Fig. 11. Abundance maps by different unmixing methods and the reference in the Cuprite dataset.

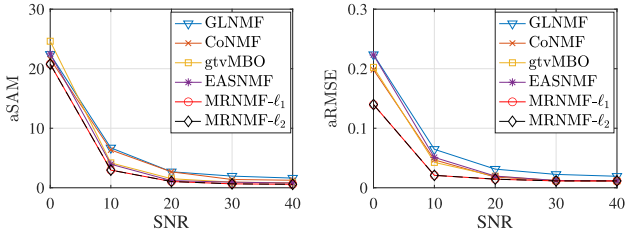


Fig. 12. Performance of HU algorithms for the synthetic datasets with Gaussian noise at different noise levels.

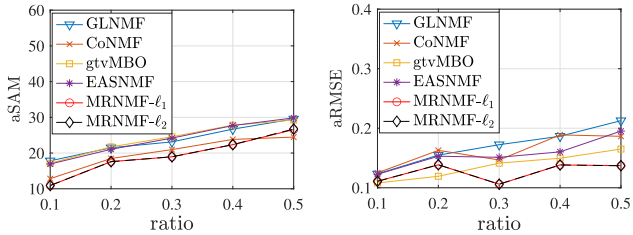


Fig. 13. Performance of HU algorithms for the synthetic datasets with sparse noise at different noise levels.

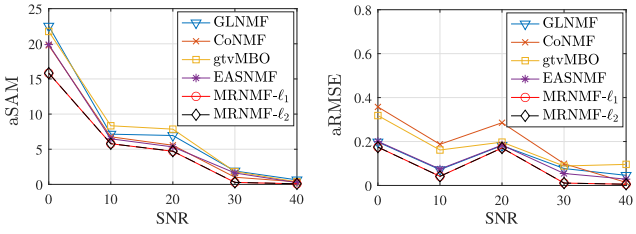


Fig. 14. Performance of HU algorithms for the Samson dataset with Gaussian noise at different noise levels.

contaminated with Gaussian and sparse noise at different noise levels, respectively, and compared with the NMF-based competing algorithms. Fig. 12 shows that both aSAM and aRMSE of these algorithms decrease as the SNR levels increase, ranging from 0 to 40 dB. From the figure, it can be seen that all the methods show some robustness to Gaussian noise, and the performance improves with increasing SNR. However, both MRNMF-ℓ_1 and MRNMF-ℓ_2 achieve the highest accuracy at different noise levels compared with other algorithms, especially at lower noise levels. This implies that the proposed MRNMF algorithm is more robust than other NMF-based methods. The similar results can also be observed in Fig. 13 for the sparse noise case, where the ratios range from 0.1 to 0.5. The ratio value of 0.1 indicates that 10% of the pixels are corrupted by the sparse noise, and refer to [26] for a more detailed description and experimental setup. Therefore, these results demonstrate the robustness of the suggested MRNMF algorithm.

In addition, the robustness of these NMF-based unmixing methods to Gaussian noise is further investigated on the real Samson and Urban datasets. In this experiment, Gaussian noise is added to the Samson and Urban datasets with different levels of 0, 10, 20, 30, and 40 dB, respectively. Fig. 14 illustrates the effect of different SNRs on the aSAM and aRMSE values of all the unmixing methods in the Samson dataset, and it can be seen that the overall performance of all the methods improves as the SNR increases, but the proposed MRNMF method performs more robustly in all the cases. As shown in Fig. 15, the robustness of all the methods to Gaussian

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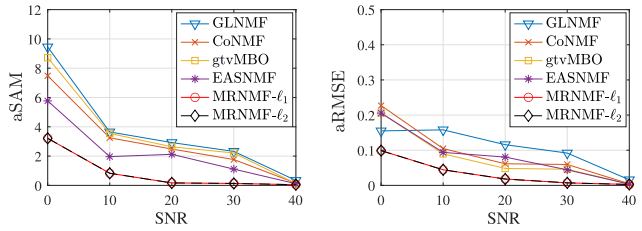


Fig. 15. Performance of HU algorithms for the Urban dataset with Gaussian noise at different noise levels.

noise on the Urban dataset still follows a similar trend to the previous datasets. Nevertheless, the performance of the proposed MRNMF method is significantly better than the other compared methods for SNRs ranging from 0 to 30 dB. The experimental results on the Samson and Urban datasets show again that the MRNMF method is very robust to higher levels of Gaussian noise.

G. Computational Complexity

Here, the computational complexity of several unmixing methods based on the NMF framework is discussed. The time taken for each iteration, including $\mathbf{A}$ and $\mathbf{X}$, as well as for the graph regularization in the early stage, is measured by counting the floating-point calculation time. The computational complexity of standard NMF and $\mathbf{X}^{-(1/2)}$ is known to be $O(mnp)$ and $O((np)^2)$, respectively. Graph Laplacian $\mathbf{L}$ of MRNMF is obtained by the graph learning Algorithm 1, and the cost mainly consists of $\mathbf{W}$ in (17) and $\mathbf{s}$ in (23). The cost of calculating $\mathbf{W}$ is $O(\sigma n)$, where σ is the scale parameter of the Gaussian filter. The linear system of $\mathbf{s}$ can be quickly solved by the conjugate gradient method, which has a computational cost of $O(n)$. Taking all the factors into consideration, the overall cost of MRNMF is $O(t(m+np)np+\sigma n)$ after t iterations. Moreover, the computational complexity of constructing the k-nearest graphs in GLNMF [31] and EASNMF [39] is $O(n^2m)$, and that of the Nyström graph in gtvMBO [32] is $O(mnq+nq^2)$, where q is the number of sampled data points. Therefore, the overall computational complexity for GLNMF and EASNMF is $O(t(m+np)np+n^2m)$, and for gtvMBO is $O(tnp(m+q)+nq(m+q))$. In addition, since the graph is not used in the CoNMF [34] method, it has an overall computational complexity of $O(t(np^2+nm^2))$.

H. Ablation Experiments

Experiments are performed using different regularizations to validate the effectiveness of MIC and RTV regularizations in HU tasks. The experiments used three datasets with reference abundance, and the results are presented in Table V. Overall, the MRNMF performance is degraded using MIC or RTV regularization alone, where $q=1$ in MIC is used as an example. The reason is that the spatial prior of the pixels in the HSI is absent with MIC regularization alone, whereas the independence prior of the endmembers in the HSI is lost with RTV regularization alone. Conversely, the combination of MIC and RTV regularization in NMF can use both prior information in the HSI, compensating for the limitations of using either prior information individually, and thus obtaining the best results.

TABLE V
ASAM AND RMSE OF ABLATION EXPERIMENTS ON THREE DATASETS

Regularization		MIC	RTV	MIC+RTV
Synthetic	aSAM	4.705	4.859	2.968
	aRMSE	0.045	0.042	0.021
	eRMSE	0.025	0.026	0.028
Samson	aSAM	8.246	8.336	2.083
	aRMSE	0.133	0.130	0.107
	eRMSE	0.080	0.082	0.065
Urban	aSAM	21.13	21.32	16.53
	aRMSE	0.265	0.263	0.163
	eRMSE	0.178	0.179	0.134

V. Conclusion

In this article, the MIC and RTV regularizations are first incorporated into the blind HU approach, and then the MRNMF algorithm is proposed based on the NMF model. The MIC regularization can effectively improve the quality of the estimated endmembers, and the RTV regularization can be adaptively learned from the original HSI and is used to constrain the estimated abundances. With the MIC, RTV, and sparsity regularizations, MRNMF can simultaneously use prior information about the endmembers and abundances and can be effectively solved using the multiplicative iteration rule. Extensive experiments are conducted to demonstrate that the proposed MRNMF is more effective and efficient compared with the competing algorithms.

Appendix

A. Proof of Theorem 1

With the concavity of the log function, the following inequality holds:

$$\log(|z|+\epsilon) \le \frac{|z|}{|u|+\epsilon} + C_1 \tag{38}$$

for any u and C_1 is a constant. The inequality of the absolute value function $|z|$ satisfies that

$$|z| \le \frac{z^2}{|u|+\epsilon} + C_2 \tag{39}$$

for any u and C_2 is a constant. Using the inequalities (38) and (39), the RTV regularization in (15) is bounded, for any vector $\mathbf{u}$, as follows:

$$\begin{aligned}
\mathrm{RTV}(\mathbf{s}) &= \sum_i \log(|(\mathbf{GDs})_i|+\epsilon) \\
&\le \sum_i \frac{|(\mathbf{GDs})_i|}{|(\mathbf{Gu})_i|+\epsilon} + C_1 \\
&\le \sum_i \frac{1}{|(\mathbf{Gu})_i|+\epsilon}(\mathbf{G}|\mathbf{Ds}|)_i + C_1 \\
&\le \sum_i \frac{1}{|(\mathbf{Gu})_i|+\epsilon}\left(\frac{\mathbf{G}(\mathbf{Ds})^{\cdot 2}}{|\mathbf{u}|+\epsilon}\right)_i + C \\
&= \sum_i \left(\mathbf{G}\frac{1}{|(\mathbf{Gu})|+\epsilon} \circ \frac{1}{|\mathbf{u}|+\epsilon}\right)_i (\mathbf{Ds})_i^2 + C
\end{aligned}$$

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where "=" holds true for $\mathbf{u} = \mathbf{Ds}$, $\mathbf{X}^{\cdot 2}$ denotes the square of the elements of the matrix $\mathbf{X}$, and C_1 and C are constants. Let $\mathbf{W}(\mathbf{u})$ be the diagonal matrix whose ith diagonal element is defined by

$$(\mathbf{W}(\mathbf{u}))_{ii} = \left(\mathbf{G} \frac{1}{|(\mathbf{Gu})| + \epsilon} \circ \frac{1}{|\mathbf{u}| + \epsilon} \right)_i \tag{40}$$

where " $\circ$ " denotes elementwise product. The matrix $\mathbf{L}(\mathbf{u})$, depending on the parameter $\mathbf{u}$, is defined as

$$\mathbf{L}(\mathbf{u}) = \mathbf{D}^T \mathbf{W}(\mathbf{u}) \mathbf{D} \tag{41}$$

which is referred to as the graph Laplacian at $\mathbf{u}$. Then, the RTV regularization satisfies the following inequality:

$$\mathrm{RTV}(\mathbf{s}) \leq \mathbf{s}^T \mathbf{L}(\mathbf{u}) \mathbf{s} + C \tag{42}$$

where "=" holds true for $\mathbf{u} = \mathbf{Ds}$. Then, the right term in the above inequality is the majorizer of $\mathrm{RTV}(\mathbf{s})$.

REFERENCES

[1] T. Kouyama, Y. Yokota, Y. Ishihara, R. Nakamura, S. Yamamoto, and T. Matsunaga, "Development of an application scheme for the SELENE/SP lunar reflectance model for radiometric calibration of hyperspectral and multispectral sensors," *Planet. Space Sci.*, vol. 124, pp. 76–83, May 2016.

[2] D. Manolakis and G. Shaw, "Detection algorithms for hyperspectral imaging applications," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 29–43, Jan. 2002.

[3] A. Brook and E. B. Dor, "Quantitative detection of settled dust over green canopy using sparse unmixing of airborne hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 2, pp. 884–897, Feb. 2016.

[4] M. A. Lee, Y. Huang, H. Yao, S. J. Thomson, and L. M. Bruce, "Determining the effects of storage on cotton and soybean leaf samples for hyperspectral analysis," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2562–2570, Jun. 2014.

[5] J. Chi and M. M. Crawford, "Spectral unmixing-based crop residue estimation using hyperspectral remote sensing data: A case study at Purdue University," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2531–2539, Jun. 2014.

[6] N. Keshava and J. F. Mustard, "Spectral unmixing," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 44–57, Jan. 2002.

[7] W. He, H. Zhang, and L. Zhang, "Total variation regularized reweighted sparse nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3909–3921, Jul. 2017.

[8] Q. Du, N. Raksuntorn, N. H. Younan, and R. L. King, "End-member extraction for hyperspectral image analysis," *Appl. Opt.*, vol. 47, no. 28, pp. 77–84, Oct. 2008.

[9] D. C. Heinz, "Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 3, pp. 529–545, Mar. 2001.

[10] X. Zhang, J. Zhang, C. Li, C. Cheng, L. Jiao, and H. Zhou, "Hybrid unmixing based on adaptive region segmentation for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 3861–3875, Jul. 2018.

[11] J. M. Bioucas-Dias et al., "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[12] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Sparse unmixing of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 6, pp. 2014–2039, Jun. 2011.

[13] J. M. Bioucas-Dias and M. Figueiredo, "Alternating direction algorithms for constrained sparse regression: Application to hyperspectral unmixing," in *Proc. 2nd Workshop Hyperspectral Image Signal Process., Evol. Remote Sens.*, Jun. 2010, pp. 1–4.

[14] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Collaborative sparse regression for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 341–354, Jan. 2014.

[15] M. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Total variation spatial regularization for sparse hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 11, pp. 4484–4502, Nov. 2012.

[16] R. Ammanouil, A. Ferrari, and C. Richard, "A graph Laplacian regularization for hyperspectral data unmixing," in *Proc. IEEE ICASSP*, Apr. 2015, pp. 1637–1641.

[17] S. Wang, T.-Z. Huang, X.-L. Zhao, G. Liu, and Y. Cheng, "Double reweighted sparse regression and graph regularization for hyperspectral unmixing," *Remote Sens.*, vol. 10, no. 7, p. 1046, Jul. 2018.

[18] T. Ince, "Double spatial graph Laplacian regularization for sparse unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[19] R. A. Borsoi et al., "Spectral variability in hyperspectral data unmixing: A comprehensive review," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 223–270, Dec. 2021.

[20] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509417.

[21] Z. Yang, M. Xu, S. Liu, H. Sheng, and H. Zheng, "Spatial-spectral attention bilateral network for hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[22] Z. Yang, M. Xu, S. Liu, H. Sheng, and J. Wan, "UST-net: A U-shaped transformer network using shifted windows for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5528815.

[23] C. Cui, X. Wang, S. Wang, L. Zhang, and Y. Zhong, "Unrolling nonnegative matrix factorization with group sparsity for blind hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5516712.

[24] R. Peharz and F. Pernkopf, "Sparse nonnegative matrix factorization with ℓ^0-constraints," *Neurocomputing*, vol. 80, pp. 38–46, Mar. 2012.

[25] X.-R. Geng, L.-Y. Ji, and K. Sun, "Non-negative matrix factorization based unmixing for principal component transformed hyperspectral data," *Frontiers Inf. Technol. Electron. Eng.*, vol. 17, no. 5, pp. 403–412, May 2016.

[26] W. He, H. Zhang, and L. Zhang, "Sparsity-regularized robust non-negative matrix factorization for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 9, pp. 4267–4279, Sep. 2016.

[27] X. Li, X. Zhang, Y. Yuan, and Y. Dong, "Adaptive relationship preserving sparse NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504516.

[28] J. Liu, Y. Zhang, Y. Liu, and C. Mu, "Hyperspectral images unmixing based on abundance constrained multi-layer KNMF," *IEEE Access*, vol. 9, pp. 91080–91090, 2021.

[29] J.-J. Wang, D.-C. Wang, T.-Z. Huang, J. Huang, X.-L. Zhao, and L.-J. Deng, "Endmember independence constrained hyperspectral unmixing via nonnegative tensor factorization," *Knowl.-Based Syst.*, vol. 216, Mar. 2021, Art. no. 106657.

[30] Y. Qian, S. Jia, J. Zhou, and A. Robles-Kelly, "Hyperspectral unmixing via $L_{1/2}$ sparsity-constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4282–4297, Nov. 2011.

[31] X. Lu, H. Wu, Y. Yuan, P. Yan, and X. Li, "Manifold regularized sparse NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 5, pp. 2815–2826, May 2013.

[32] J. Qin et al., "Blind hyperspectral unmixing based on graph total variation regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3338–3351, Apr. 2021.

[33] T. Yang, M. Song, S. Li, and Y. Wang, "Spectral–spatial anti-interference NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5524817.

[34] J. Li, J. M. Bioucas-Dias, A. Plaza, and L. Liu, "Robust collaborative nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6076–6090, Oct. 2016.

[35] J. M. P. Nascimento and J. M. Bioucas-Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.

[36] L. Zhuang, C. Lin, M. A. T. Figueiredo, and J. M. Bioucas-Dias, "Regularization parameter selection in minimum volume hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9858–9877, Dec. 2019.

[37] J. Wang and C.-I. Chang, "Applications of independent component analysis in endmember extraction and abundance quantification for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 9, pp. 2601–2616, Sep. 2006.

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[38] E. M. M. B. Ekanayake et al., "Constrained nonnegative matrix factorization for blind hyperspectral unmixing incorporating endmember independence," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11853–11869, 2021.

[39] J. Zhang, X. Zhang, and L. Jiao, "Sparse nonnegative matrix factorization for hyperspectral unmixing based on endmember independence and spatial weighted abundance," *Remote Sens.*, vol. 13, no. 12, p. 2348, Jun. 2021.

[40] Z. Wei, J. Zhang, Z. Xu, Y. Liu, and K. Okarma, "Measurement matrix optimization via mutual coherence minimization for compressively sensed signals reconstruction," *Math. Problems Eng.*, vol. 2020, pp. 1–18, Sep. 2020.

[41] D. Barchiesi and M. D. Plumbley, "Learning incoherent dictionaries for sparse approximation using iterative projections and rotations," *IEEE Trans. Signal Process.*, vol. 61, no. 8, pp. 2055–2065, Apr. 2013.

[42] D. D. Lee and H. S. Seung, "Algorithms for non-negative matrix factorization," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 13, Feb. 2001, pp. 556–562.

[43] E. J. Candès, J. K. Romberg, and T. Tao, "Stable signal recovery from incomplete and inaccurate measurements," *Commun. Pure Appl. Math.*, vol. 59, no. 8, pp. 1207–1223, Aug. 2006.

[44] L. Xu, Q. Yan, Y. Xia, and J. Jia, "Structure extraction from texture via relative total variation," *ACM Trans. Graph.*, vol. 31, no. 6, pp. 1–10, 2012.

[45] Y. Sun, P. Babu, and D. P. Palomar, "Majorization-minimization algorithms in signal processing, communications, and machine learning," *IEEE Trans. Signal Process.*, vol. 65, no. 3, pp. 794–816, Feb. 2017.

[46] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.

[47] C.-I. Chang and Q. Du, "Estimation of number of spectrally distinct signal sources in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 3, pp. 608–619, Mar. 2004.

[48] J. M. Bioucas-Dias and J. M. P. Nascimento, "Hyperspectral subspace identification," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 8, pp. 2435–2445, Aug. 2008.

[49] R. N. Clark, G. A. Swayze, R. Wise. (2007). *USGS Digital Spectral Library Splib06a: US Geological Survey, Digital Data Series 231*. [Online]. Available: http://speclab.cr.usgs.gov/spectral.lib06

[50] H. Li, R. Feng, L. Wang, Y. Zhong, and L. Zhang, "Superpixel-based reweighted low-rank and total variation sparse unmixing for hyperspectral remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 629–647, Jan. 2021.

[51] R. A. Borsoi, T. Imbiriba, J. C. M. Bermudez, and C. Richard, "A fast multiscale spatial regularization for sparse hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 4, pp. 598–602, Apr. 2019.

[52] T. Ince, "Superpixel-based graph Laplacian regularization for sparse hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[53] F. Zhu, "Hyperspectral unmixing: Ground truth labeling, datasets, benchmark performances and survey," 2017, *arXiv:1708.05125*.

[54] D. Hong and X. X. Zhu, "SULoRA: Subspace unmixing with low-rank attribute embedding for hyperspectral data analysis," *IEEE J. Sel. Topics Signal Process.*, vol. 12, no. 6, pp. 1351–1363, Dec. 2018.

[55] C. Y. Zheng, H. Li, Q. Wang, and C. L. Philip Chen, "Reweighted sparse regression for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 479–488, Jan. 2016.

[56] R. Wang, H. Li, A. Pizurica, J. Li, A. Plaza, and W. J. Emery, "Hyperspectral unmixing using double reweighted sparse regression and total variation," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 7, pp. 1146–1150, Jul. 2017.



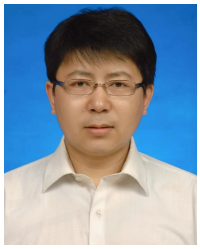
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